

A little bit about the course!

Assumed Knowledge

- Basic Linear Algebra
 - System of Linear Equations $\mathbf{A}\vec{x} = \vec{C}$
 - System of nonlinear equations $\vec{f}(\vec{x}) = 0$
- Basic Calculus
 - Taylor series
 - Differentiation/Integration
 - Multivariate calculus
 - Differential Equations
- Basic Programming
 - Knowledge of at least 1 programming language (Python, R, Java, C, C++, etc)
 - Conditional (if/else) Statements
 - For/while loops
 - Arrays
 - Functions

Assessment

- Assignment 1 (20%)
- Assignment 2 (30%)
- Final Exam (40%)
 - Have access to your laptop
 - No internet.
 - Similar to exercises (see below).
- Attendance and Discussion (10%)
 - Submit homework exercises at the end of the 3 hour session
 - Start from Day 2, last one on Day 14 (Day 15 is the exam). Each one is worth 1%. Maximum is 10%.

Delivery

- Tour of the Canvas Site
- Material will be delivered using Jupyter Notebooks and slides.
- Material will be available before the next class.
- Material will be delivered using slides, blackboard and live coding.

1.1 Introduction and Motivation

- Why find the minimum (or maximum) of a function?
- Why Julia?
- Why Scientific Machine Learning?

Why do we need to find the minimum (or maximum) of a function?

There are many engineering applications where we need to find the minimum of a function. For example

- Minimize the material needed to build a computer to reduce weight and cost.
- Minimum thickness of a beam so that it can support a load on a building
- Minimizing time to transport assets/goods to a particular destination to save cost.
- In machine learning, you want to minimize the cost function so that you can get the best possible mathematical model for accurate predictions of an engineering system.

The area of the cylinder is given by

$$A(r) = \pi r^2 + \pi r^2 + 2\pi r h$$

You are given that the volume is 1

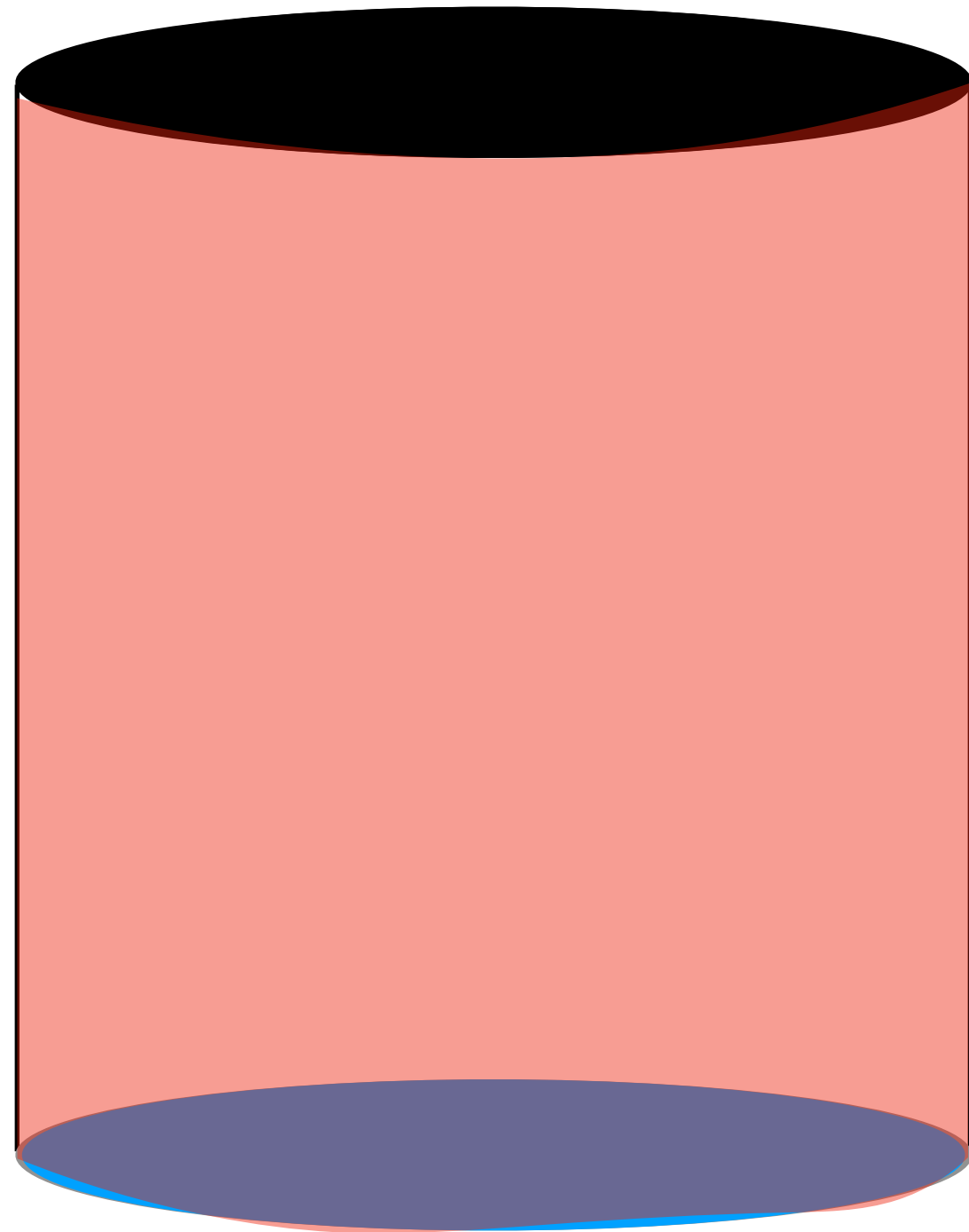
$$V = \pi r^2 h = 1$$

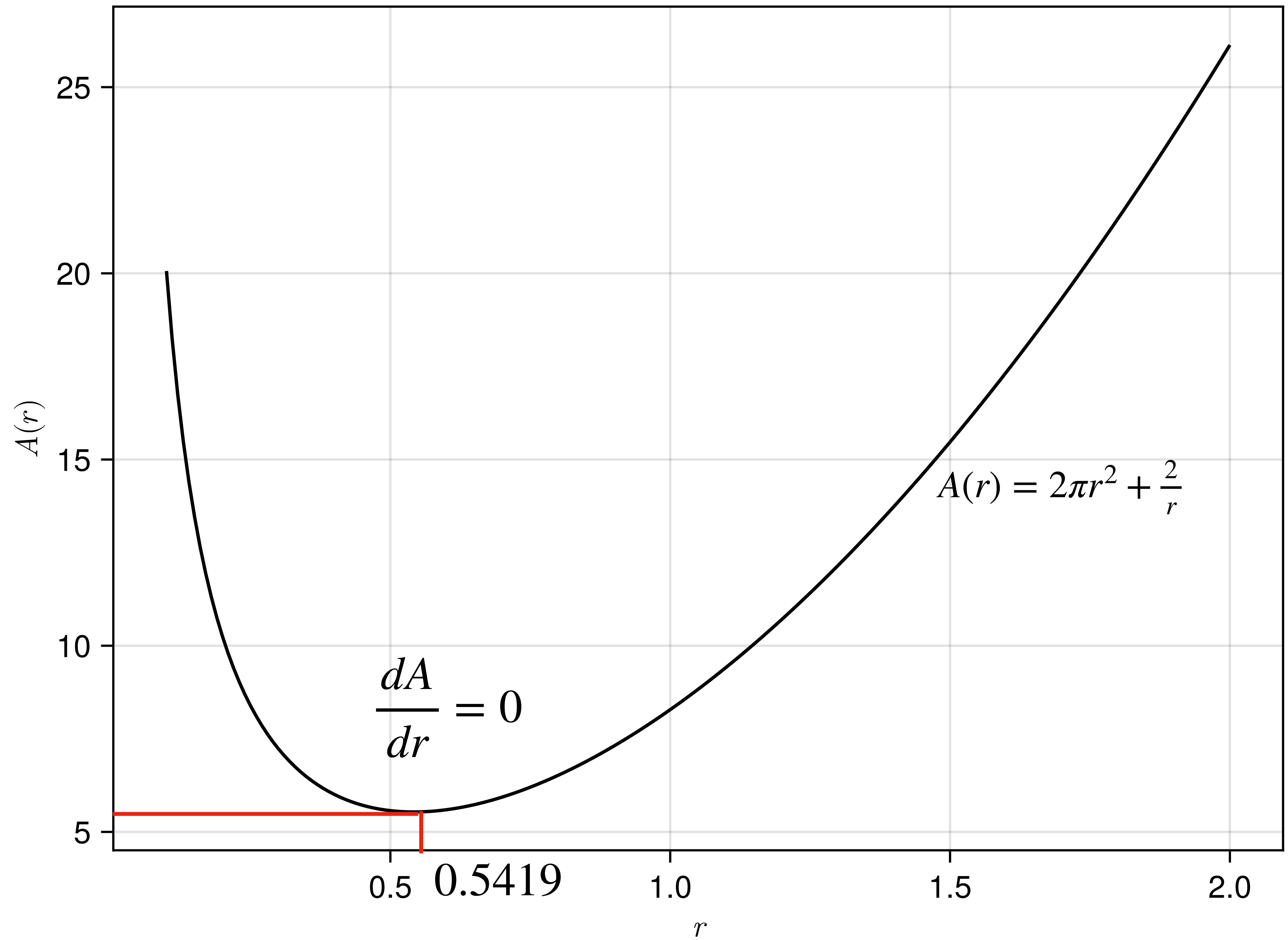
$$h = \frac{1}{\pi r^2}$$

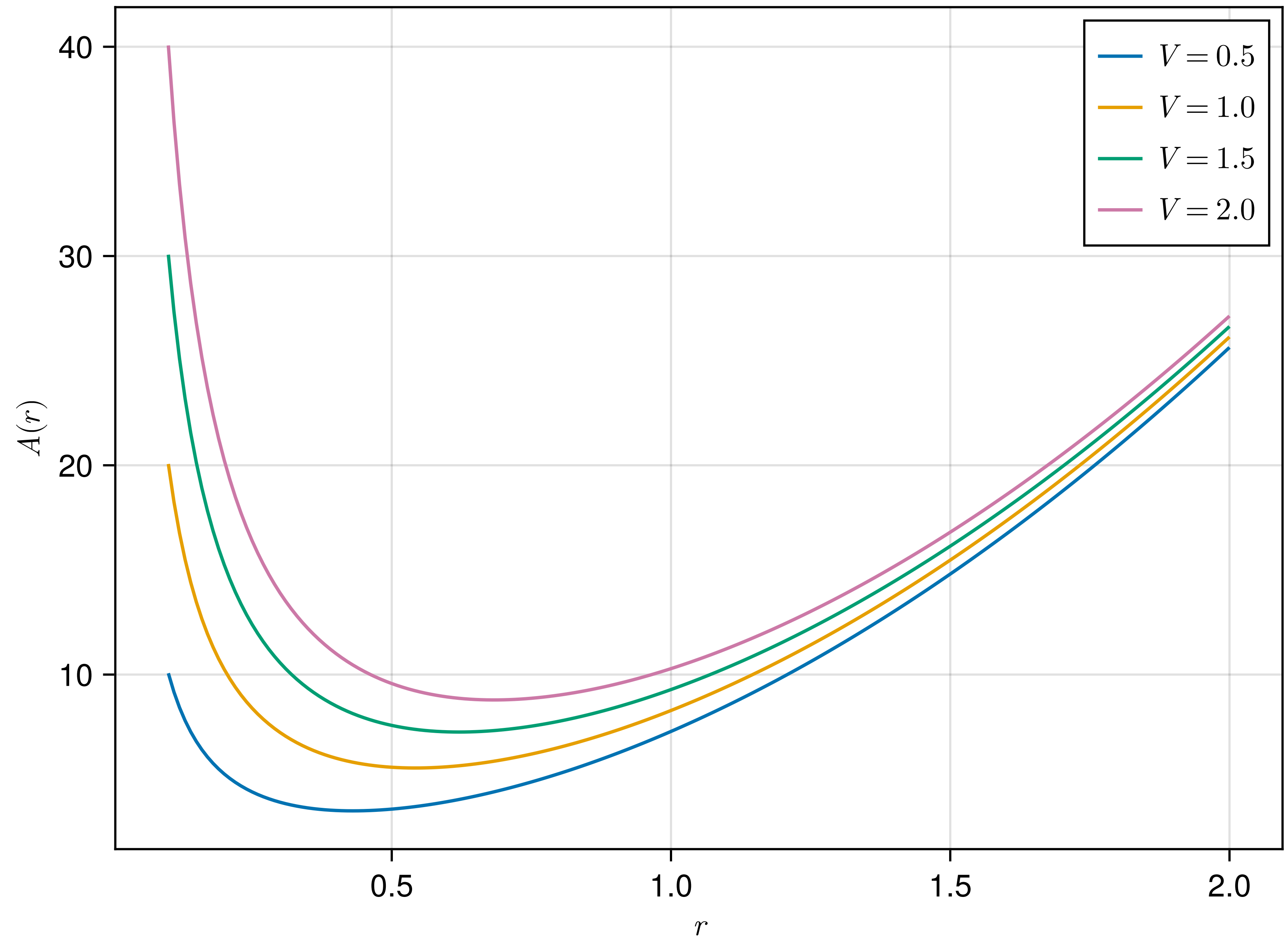
Substitute into the expression for $A(r)$ gives

$$A(r) = \pi r^2 + \pi r^2 + 2\pi r \frac{1}{\pi r^2}$$

$$A(r) = 2\pi r^2 + \frac{2}{r}$$







Why Julia?

Why Julia?

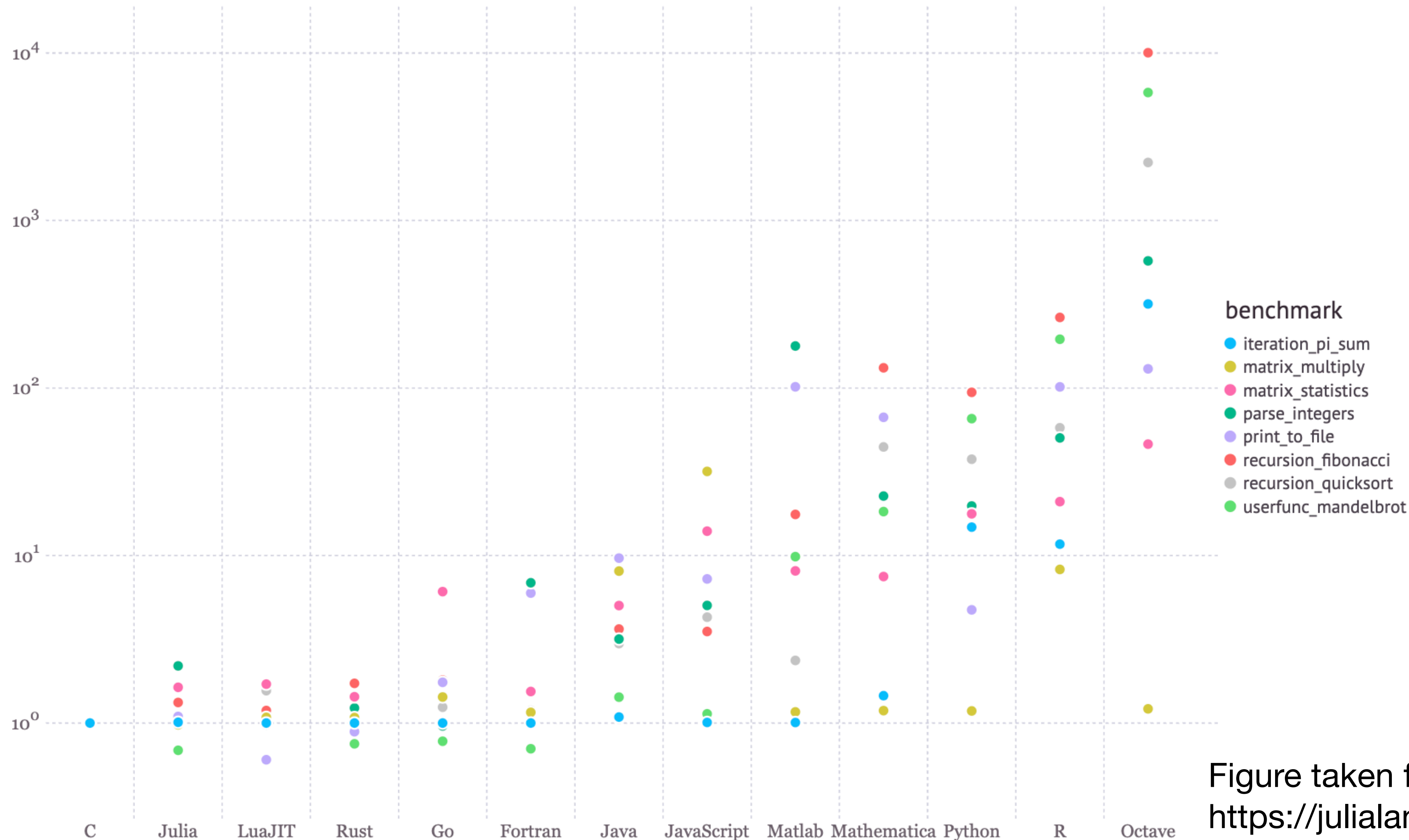
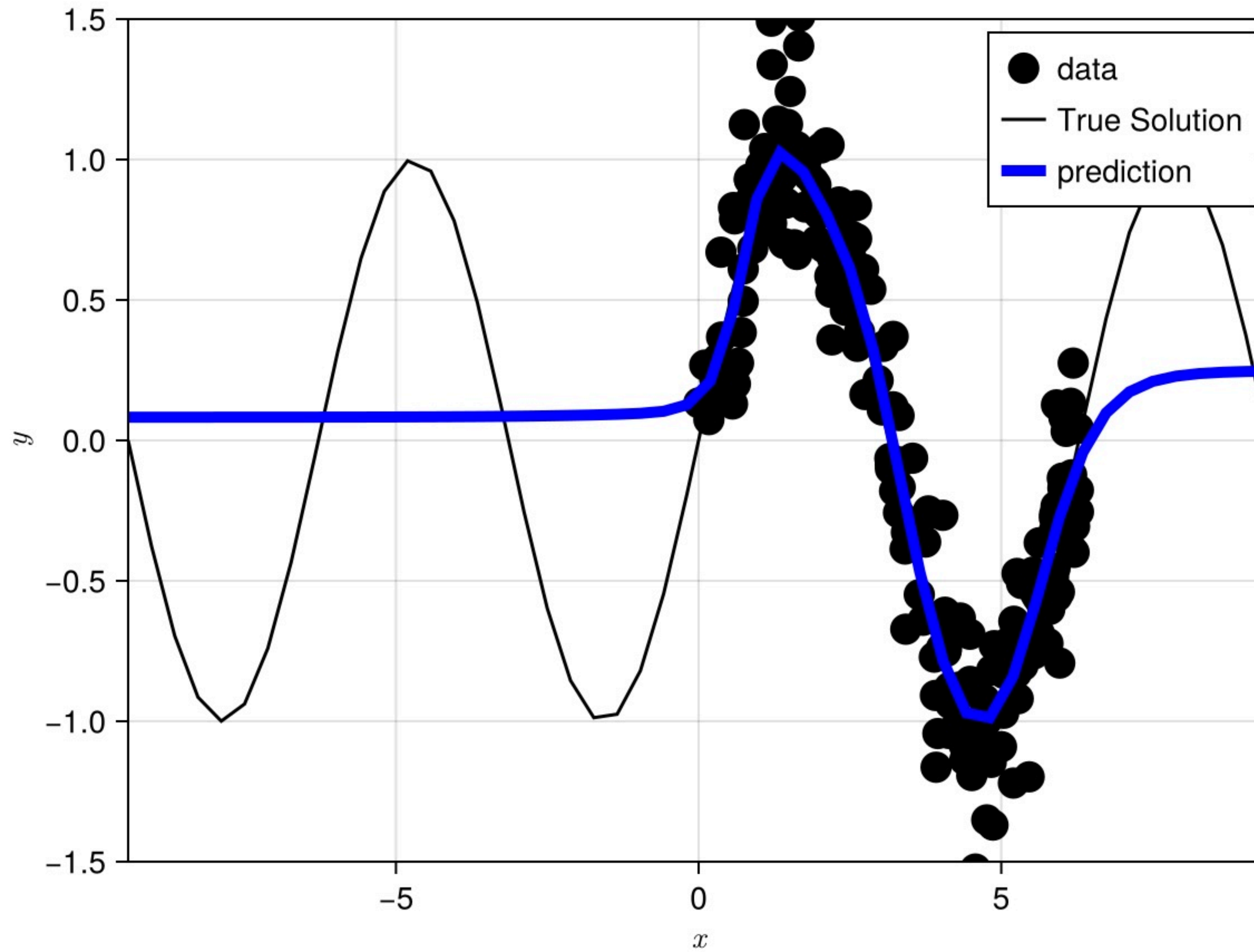


Figure taken from
<https://julialang.org/benchmarks/>

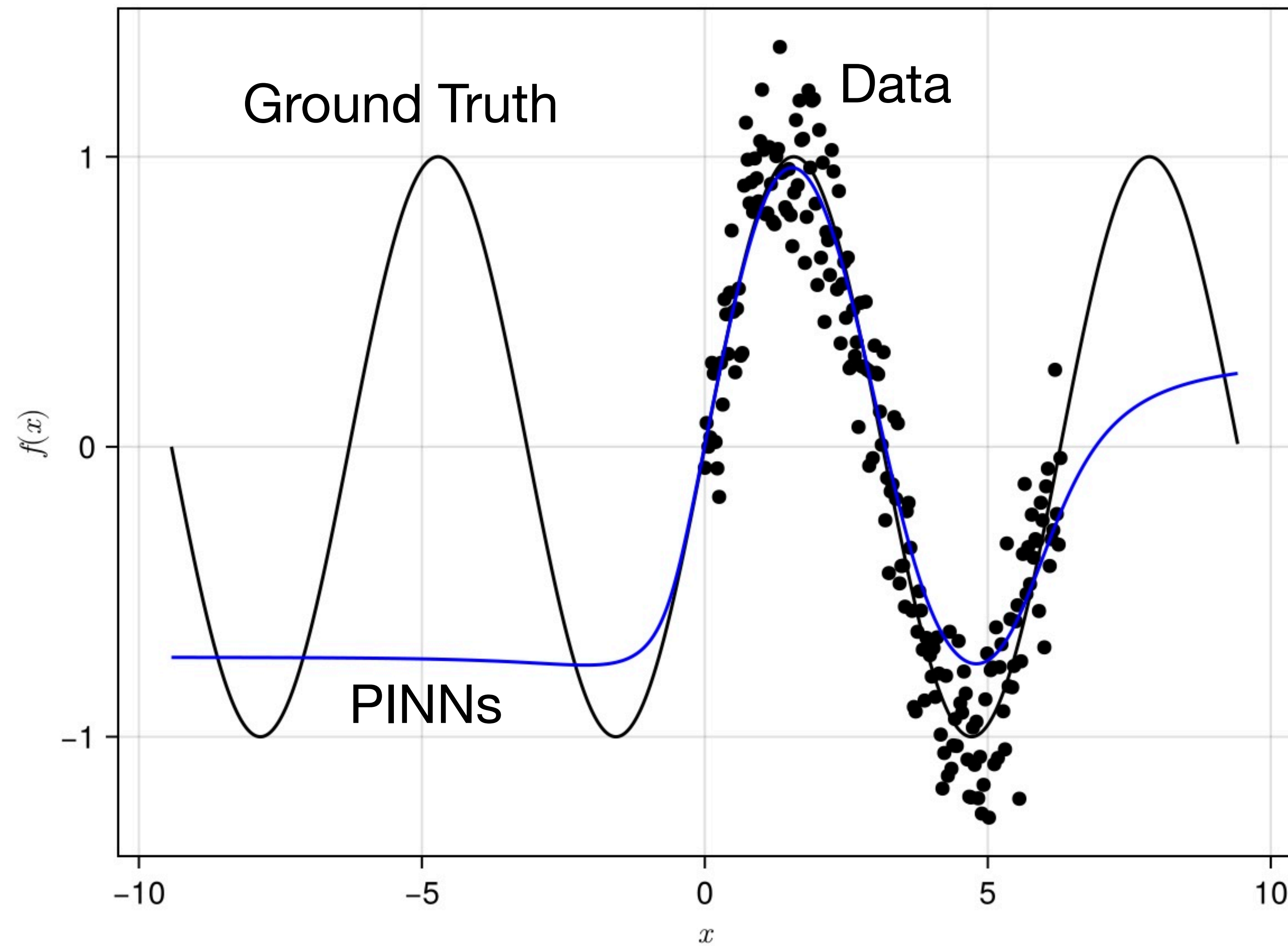
Why Scientific Machine Learning?

Why Scientific Machine Learning?

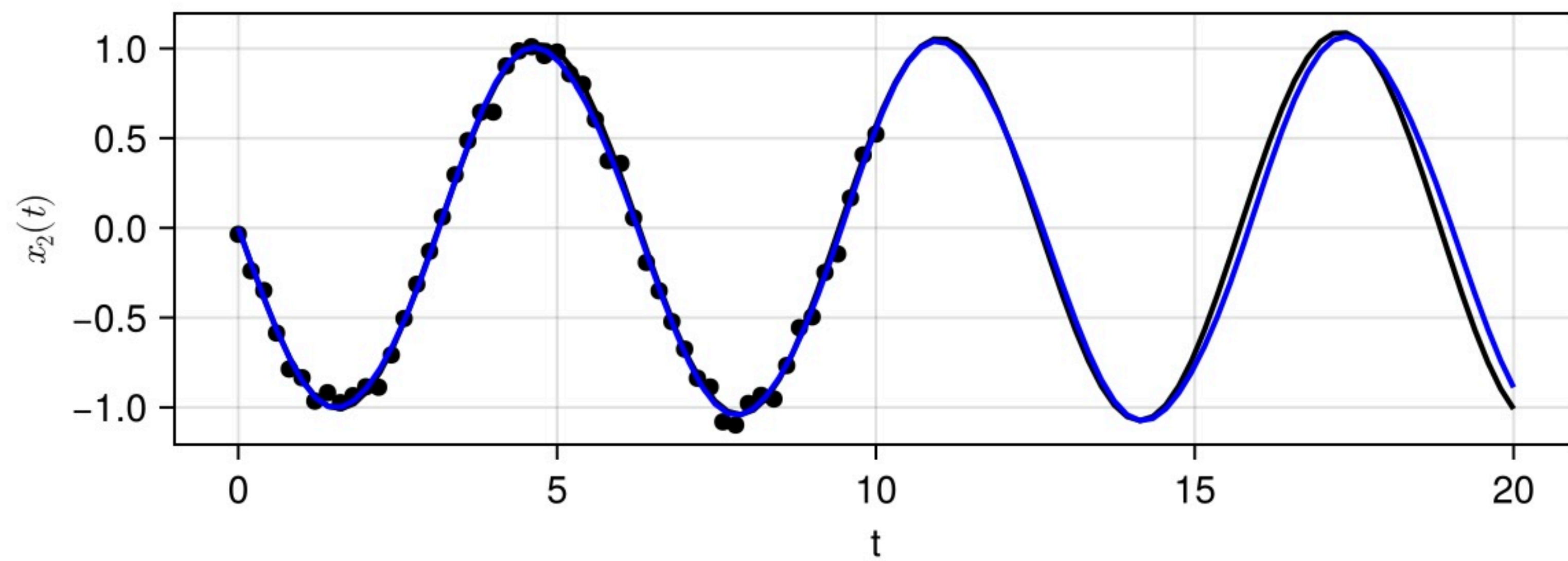
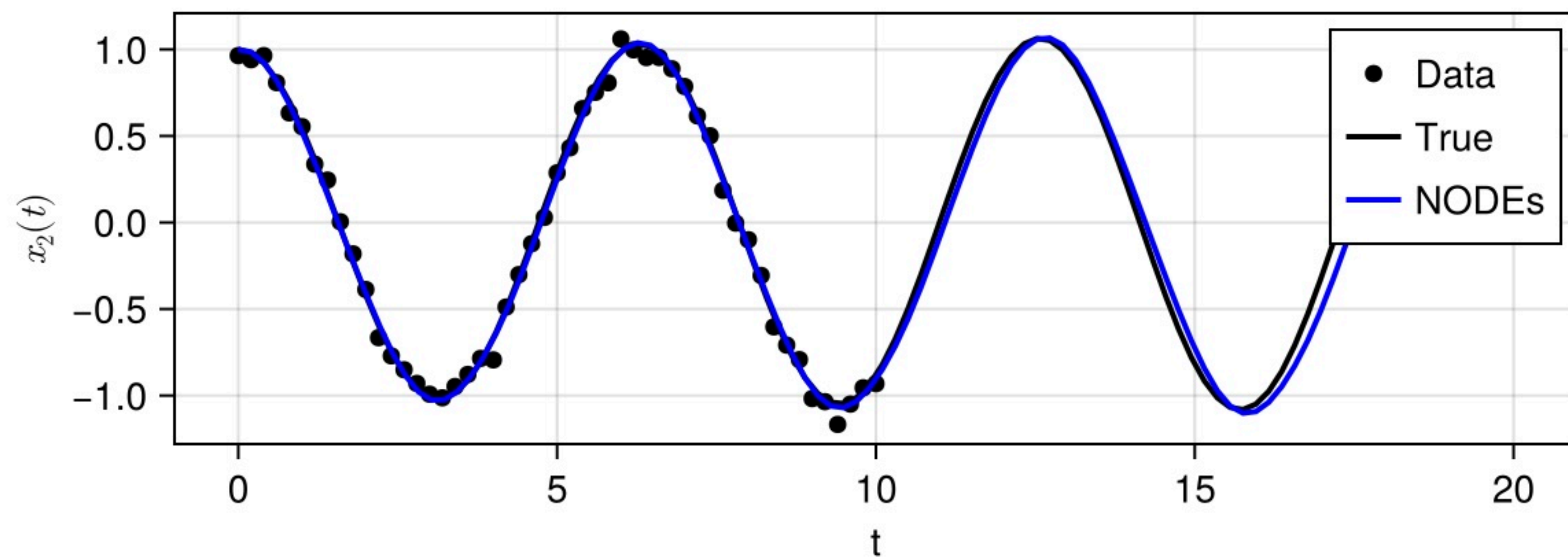
- Incorporate Physical “constraints” into the predictions....not just rely on data
- Scientific applications are usually “data poor”. Need to incorporate physical knowledge (conservation laws) to obtain better predictions



Traditional
Machine
Learning



PINNs
(Physics Informed
Neural Network)



Install Julia!

- **Instructions on Canvas**

- File -> Open Folder
- New Folder -> Folder name
- Julia: Start REPL
- In Julia REPL, type `pwd()` to find confirm which directory you are in.
- Press `]` to enter the package manager
- Activate `.` to activate the current folder
- Add “package-name” to add the package
-

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